

Mall Customer Segmentation using K-Means Clustering

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Introduction

Customer segmentation is the process of dividing customers into groups based on similar characteristics. This project uses K-Means Clustering to segment mall customers according to their annual income and spending score.

Objective

To identify different groups of customers based on their purchasing behavior and income levels.

Dataset Description

Dataset Name: Mall Customers Dataset.

Attributes: CustomerID, Gender, Age, Annual Income (k\$), Spending Score (1-100).

Algorithm Used

K-Means Clustering.

Methodology

1. Load Dataset
2. Data Exploration
3. Feature Selection
4. Elbow Method
5. Apply K-Means Clustering
6. Visualize Clusters
7. Analyze Results

Results

Five customer segments were identified based on annual income and spending score. These segments help businesses understand customer behavior and improve marketing decisions.

Conclusion

K-Means Clustering successfully segmented customers into meaningful groups. The project demonstrates how machine learning can be used for customer analysis and targeted marketing.